

DEFENSE HEALTH INTELLIGENCE PLATFORM

WelfareAI System Handbook

Complete End-to-End Operational Guide & Architecture Reference

A comprehensive, easy-to-understand breakdown of every screen, algorithm, database pipeline, and workflow across all three user roles: **Personnel (Soldier)**, **Welfare Officer**, and **System Administrator**.

Project Title:	AI-Based Personnel Stress & Welfare Monitoring System
Problem Statement:	SIH 2026 - Defense & Paramilitary Welfare
Technology Stack:	Node.js / Express, Python / FastAPI, Scikit-Learn, Vanilla JS
Document Classification:	Operational & Technical Handbook
System Version:	v1.4.0 (Production Verified)
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1. Executive Summary & Why This Project Exists

1.1 The Operational Problem in Armed Forces

In military and paramilitary organizations (such as CRPF, BSF, ITBP, and Indian Army), service members operate under extreme physical and mental stress:

- **Harsh Deployments:** Extreme cold, high altitudes, remote jungle warfare, and anti-insurgency combat.
- **Severe Shift Demands:** Continuous night watch duties (often 56+ hours per week) disrupting biological sleep cycles.
- **Family & Geographic Separation:** Prolonged isolation from home with limited contact during active deployments.
- **The Mental Health Stigma:** Soldiers rarely complain or seek help voluntarily out of fear of career impact, peer judgment, or administrative discrimination.

As a result, psychological strain remains invisible until it accumulates into severe chronic burnout, clinical depression, accidental weapon mishandling, or tragic suicides.

1.2 The WelfareAI Solution

WelfareAI transforms defense welfare from a *reactive crisis response* into a **proactive, data-driven early warning radar**. By combining scientifically validated psychometric assessments (Perceived Stress Scale - PSS-10) with physiological, sleep, and operational duty factors, our machine learning model detects rising strain weeks before a crisis occurs—while upholding strict privacy safeguards and human officer oversight.

Key Philosophy — "Human-in-the-Loop AI":

The AI never replaces commanders or doctors. It acts as an intelligent early-warning stethoscope that highlights invisible strain, calculates duty-rebalancing rotations, and empowers officers to provide timely human care.

2. System Architecture & The Three User Roles

2.1 High-Level Architecture Flow

The platform is designed as a secure, distributed microservice architecture with three clear tiers:

[1. FRONTEND TIER (Vanilla HTML5 / CSS3 / JavaScript)] • 24 Responsive Web Portals (Zero heavy frameworks, blazing fast, works on low-bandwidth field tablets) • Dynamic Role-Based Access Control (Personnel, Officer, Admin) | ▼ HTTPS / REST APIs [2. BACKEND API TIER (Node.js & Express - Port 5000)] • JWT Authentication & Bcrypt Password Hashing • RBAC Middleware & Security Audit Logging • Persistent File-Based / MongoDB Hybrid Database Adapter | ▼ Internal Microservice Calls [3. MACHINE LEARNING TIER (Python & FastAPI - Port 8000)] • Trained Random Forest / LightGBM Classifier (81.5% Evaluated Accuracy) • Real-Time Risk Score Calculator (0 - 100%) • Feature Attribution Engine (Multi-Factor Breakdown & Action Item Generator)

2.2 The Three User Roles & Permission Matrix

Role	Who Uses It?	Core Capabilities	Restricted From
1. Personnel (Soldier)	Constable, Havildar, Sub-Inspector, Field Soldiers	• Submit periodic confidential check-ins • View personal AI risk score & explainable factor breakdown • Receive personalized wellness guidance • Track longitudinal recovery trajectory • Submit confidential support requests	Cannot view other soldiers' records, cannot access command dashboards, cannot change shift rosters.
2. Welfare Officer	Battalion Welfare Officer, Assistant Commandant, Medical Liaison	• Triage high-strain alerts across the entire battalion • View the 4-Quadrant Early-Warning Radar • Access Authorized Unit Personnel Roster • Approve AI Shift Pacing & Rest Rotations (Adjutant Desk) • Track closed-loop follow-up recovery velocities • Resolve confidential soldier support requests	Cannot alter system configurations, cannot create/delete officer or admin accounts.
3. System Administrator	HQ IT Directorate, Cyber Compliance Officer	• User Governance (Activate/suspend soldiers and officers) • ML Model Transparency & Live Accuracy Verification • System Telemetry &	Cannot view personal psychological questionnaires of soldiers (strictly protected by privacy walls).

Role	Who Uses It?	Core Capabilities	Restricted From
		Component Health Status • Immutable Security Audit Trail Logs	

3. Public & Shared Modules Walkthrough

3.1 Public Portal & Landing (index.html / landing.html)

The public gateway of the platform. It presents the national mission of the project: safeguarding defense and paramilitary wellness through AI intelligence.

- **Key Highlights:** Overview of the platform's multi-factor monitoring, zero-stigma confidential design, and defense-grade encryption standards.
- **Direct Navigation:** Provides one-click access to Secure Login, Registration, Public Project Dossier, and Help Hotline directory.

3.2 Secure Login (login.html)

The central security authentication point for all three roles.

- **How it Works:** Users enter their Service ID / Email and Password.
- **Smart Role-Based Redirection:**
 - Soldiers (**PERSONNEL**) → Redirected to dashboard.html
 - Welfare Officers (**WELFARE_OFFICER**) → Redirected to welfare-officer.html
 - Administrators (**ADMIN**) → Redirected to admin.html
- **Security Standards:** Generates a cryptographically signed JWT token stored securely, validated against every API request.

3.3 Personnel Registration (register.html / signup.html)

Enables onboarding of military service members into the battalion database.

- **Fields Captured:** Full Name, Military Service Number (e.g. CRPF-9042), Rank (Havildar, Sepoy, etc.), Unit Designation (e.g. *CRPF Battalion 104*), Contact details, and secure password.
- **Data Validation:** Enforces strong passwords and verifies unique Service Number registration to prevent duplicate records.

3.4 Project Dossier (project-dossier.html)

A comprehensive architectural and technical showcase intended for command review and technical judges.

- **Contents:** Explains the mathematical formulations (PSS-10, composite risk calculations), machine learning training pipeline, data protection protocols, and API endpoint documentation.

3.5 Privacy & Safeguards (privacy.html)

Details the strict data ethics protocols protecting service members.

- **Guarantees:** Role-based access control, medical confidentiality, encrypted data transmission, and non-punitive welfare assurance (mental health assessments can never be weaponized for negative career appraisals).

3.6 Help & Emergency Directory (help.html)

A 24/7 lifeline directory providing immediate access to toll-free defense helplines (e.g., *Forces Tele-Counseling Line 1800-180-4024*), base medical officer extensions, and platform usage FAQs.

4. Personnel Role (Soldier) — Screen-by-Screen Walkthrough

4.1 Personnel Dashboard (dashboard.html)

The home screen when a soldier logs in. Designed with clean, calming visual aesthetics to reduce anxiety.

- **Current Welfare Status Card:** Displays the soldier's live status (**LOW / BALANCED** , **MODERATE STRAIN** , or **HIGH WELFARE CONCERN**) alongside their Composite Risk Score (e.g., 42%).
- **Quick Check-in Prompt:** Informs the soldier when their next assessment is due (e.g., "Last check-in was 2 days ago").
- **Weekly Trajectory Sparkline:** A compact line graph showing their recent wellness trend.
- **Targeted Action Items:** Displays up to 3 immediate personal recovery tips (e.g., sleep reset, hydration, physical relaxation).

4.2 Confidential Stress Check-in (stress-checkin.html)

The core diagnostic tool where the soldier completes their periodic check-in in under 3 minutes.

- **Section A: Psychometric Scale (PSS-10):** 10 standardized clinical questions measuring perceived helplessness, unpredictability, and overload over the past month.
- **Section B: Physiological & Environmental Factors:** Captures daily sleep duration (hours), weekly duty hours (e.g. 56h/wk), physical exhaustion level, and family contact frequency.
- **Section C: Audio / Voice Sentiment (Optional):** Allows the soldier to record a 15-second voice sample. The backend evaluates acoustic tone, pitch jitter, and cadence as secondary strain indicators.
- **Instant Submission:** Sends payload to `/api/v1/checkin`, triggering the Python FastAPI ML microservice for real-time inference.

4.3 Instant AI Assessment Result (ai-analysis.html)

Immediately loaded after submitting a check-in. Presents the machine learning diagnostic results in clear, non-judgmental language.

- **Composite Risk Gauge:** Visual circular gauge displaying risk percentage (0 to 100%).
- **Multi-Factor Stress Radar:** A visual radar chart plotting strain across 5 dimensions: Workload, Sleep Deprivation, Emotional Helplessness, Environmental Load, and Recovery Deficit.
- **Factor Influence Breakdown:** Explains exactly which factor contributed most to their score (e.g., "Sleep deprivation contributed +38% to your current risk").

4.4 Personalized Recommendations (recommendations.html)

Translates diagnostic AI output into actionable personal welfare habits.

- **Tailored Interventions:** Divided into Sleep Hygiene, Cognitive Reset, Physical Decompression, and Social Connection.
- **Interactive Checklists:** Soldiers can mark action items as "Completed" to track their own wellness progress.

4.5 Longitudinal Trends & Trajectory (trends.html)

Visualizes the soldier's mental health trajectory across weeks and months.

- **Welfare Index vs. Normalized PSS Graph:** Displays two comparative curves over time: the AI-predicted welfare score vs. their raw psychological questionnaire responses.
- **Total Recorded Check-ins:** Total count of assessments completed.
- **Average Risk Score:** The running mathematical mean of their mental health index.
- **Trajectory Velocity:** The rate of change between recent check-ins (e.g., Improving -6% or Stable).

4.6 Recovery Journey Tracking (recovery-journey.html)

Gamified recovery roadmap showing a soldier's step-by-step progress from a high-strain state back to an optimal, balanced baseline. Celebrates wellness milestones and sustained recovery streaks.

4.7 Explainable AI — "Why Result" & "Why Risk High" (why-result.html / why-risk-high.html)

Transparency screens that demystify the machine learning model so the soldier never feels "judged" by an opaque black box.

- **Feature Attribution Table:** Lists every input feature (duty hours, sleep, PSS score) alongside its exact percentage impact on the final risk calculation.
- **Transparent Baseline:** Shows how their current inputs compare against standard healthy military baselines.

4.8 Check-in Delta Analysis (what-changed.html)

A comparative side-by-side breakdown between the soldier's **previous check-in** and their **current check-in**.

- Highlights changes in red and green (e.g., *Sleep increased from 4.5 hrs to 6.8 hrs (+51%)* \$ ightarrow\$ *Risk dropped by -22%*).

4.9 Interactive Sensitivity Sandbox (explanation-lab.html)

An interactive "What-If" simulator allowing soldiers to explore how lifestyle changes affect their risk score.

- Soldiers adjust sliders (e.g., increasing sleep hours, reducing caffeine, adjusting duty breaks) and watch the simulated risk gauge drop in real-time, encouraging healthy behavioral modifications.

4.10 Confidential Support Requests (support.html)

The confidential lifeline where soldiers submit private requests directly to the Unit Welfare Officer.

- **Request Form:** Soldier selects category (*Confidential Counseling, Duty Rotation, Medical / Health, Family Emergency*), sets urgency (*Low, Priority, Urgent*), and writes a brief note.
- **My Submitted Requests:** Displays all submitted requests, live status (PENDING REVIEW \$ ightarrow\$ RESOLVED), and the **Officer Resolution Notes** explaining the appointment or assistance provided.

4.11 Profile & Settings (profile.html / settings.html)

Allows service members to manage their personal profile, review emergency contacts, configure notification preferences, and toggle dark/light theme display.

5. Welfare Officer Role — Master Command Portal Walkthrough

The Welfare Officer acts as the battalion's clinical and operational guardian. Their entire operational command is centralized in `welfare-officer.html` across 6 interconnected tabs, supported by the dedicated `early-warning.html` radar.

5.1 Top Executive Summary Cards (Always Visible)

At the top of every tab, the Welfare Officer has real-time visibility over the battalion's macro posture:

Metric	What It Tracks	Operational Meaning
Monitored Personnel	Total registered soldiers in the unit roll.	Guarantees 100% census coverage.
High Welfare Signals	Count of personnel whose AI risk score exceeds 65%.	Immediate triage count requiring clinical attention.
Pending AI Alerts	Unacknowledged alerts waiting in the triage inbox.	Officer's active action queue.
Active Follow-ups	Soldiers currently undergoing scheduled re-assessment.	Ensures closed-loop recovery tracking.

5.2 TAB 1: AI Triage & Alerts

The primary clinical inbox. Whenever a soldier completes a check-in resulting in a High Risk score ($> 65\%$) or expressing severe distress markers, an automated alert is generated.

- Alert Information:** Displays soldier's Name, Rank, Service ID, Risk Score, Date, and Top Contributing Stress Factors.
- Human-in-the-Loop Actions:**
 - Review & Add Notes:** Officer records medical/clinical observations.
 - Assign Follow-up:** Toggles a mandatory 7-day or 14-day re-assessment schedule.
 - Acknowledge / Resolve:** Formally moves alert from OPEN to ACKNOWLEDGED or RESOLVED.

5.3 TAB 2: Early-Warning Center & Dedicated Radar (`early-warning.html`)

A dynamic early-warning scatter matrix that classifies all battalion personnel into **4 Operational Quadrants** based on their risk score and trajectory velocity:

Quadrant	Classification Criteria	Recommended Command Action
 New Signals	First-time elevated score with no historical baseline.	Early check-in outreach; monitor next cycle.

Quadrant	Classification Criteria	Recommended Command Action
 Rising Signals	Risk score accelerating upward by $\geq +6\%$ since last check-in.	Proactive intervention to arrest velocity before critical burnout.
 Persistent Signals	Chronically high strain across 2 or more consecutive cycles.	Mandatory medical consultation and duty schedule rotation.
 Improving Signals	Risk delta decreasing by $\leq -6\%$ following intervention.	Verify recovery stability; reinforce positive progress.

5.4 TAB 3: Authorized Unit Personnel Roster

The master operational roll of all military personnel assigned under this unit's jurisdiction (e.g., *CRPF Battalion 104*).

- **Table Columns:**

- **Personnel ID:** Official military serial (e.g. CRPF-9042).
- **Full Name:** Soldier's legal name.
- **Rank:** Military rank (e.g. *Havildar, Sepoy*).
- **Unit:** Active company / battalion deployment.
- **Latest Signal:** Color-coded AI welfare badge (**LOW BASELINE** , **MODERATE STRAIN** , **HIGH WELFARE CONCERN**).
- **Last Check-in:** Exact timestamp of last assessment or Never / Unassessed.

- **Export Tools:** Top-right buttons generate instant **Export CSV** files and **Executive PDF Dossiers** for morning command briefings with the Commanding Officer (CO) and Adjutant.

5.5 TAB 4: AI Shift Pacing & Rest Rotation Optimizer

The bridge between psychological telemetry and operational military scheduling.

- **How it Works:** Identifies high-strain personnel ($\$Risk > 50\%$) whose fatigue is driven by strenuous assignments (such as 56h/wk High-Altitude Night Perimeter Watch).
- **The Recommendation Card:**
 - Displays *Current Duty* vs. *AI Recommended Pacing Reallocation* (e.g., Day Base Communications & Logistics for 72 Hours).
 - Predicts specific strain relief (e.g., **Predicted Relief: -34%**).
 - Calculates estimated battalion-wide fatigue reduction (e.g., **~26.4%**).
 - Generates a formal **Adjutant Protocol Reference** (e.g., PROP-CRPF-9042-801).
- **Approve Pacing Reallocation:** 1-click authorization dispatches the schedule change to the Adjutant's duty desk and logs an immutable security audit entry.

5.6 TAB 5: Follow-up & Re-Analysis Tracking Workflow

Scientifically measures recovery velocity across check-in cycles to close the care loop.

- Displays scheduled follow-up dates, the soldier's **Initial Risk Index** (e.g., 72%), their new **Re-analyzed Index** (e.g., 38%), and the resulting trajectory delta (**Improved -34%**).
- If no follow-ups are currently pending, it clearly notifies the officer: "*No follow-up sessions currently scheduled.*"

5.7 TAB 6: Personnel Support Requests Queue

The officer's confidential inbox for soldier help requests.

- Shows the soldier's name, category (e.g., *Confidential Counseling*), and urgency level.

- Clicking "**Update / Respond**" opens a prompt where the officer enters official resolution notes (e.g., appointment time, leave approval).
- Once submitted, the request is marked RESOLVED, and an in-app notification is immediately delivered to the soldier's portal.

6. System Administrator Role — Master Admin Portal Walkthrough

The System Administrator oversees governance, system health, machine learning compliance, and security auditing via `admin.html`.

6.1 User Governance & Role Management

Provides complete lifecycle management over user accounts across the platform:

- **User Directory Table:** Lists all registered users with their Name, Email, Service Number, Role (**PERSONNEL**, **WELFARE_OFFICER**, **ADMIN**), and Account Status.
- **Administrative Controls:** Ability to activate, suspend, or update account roles, ensuring only authorized personnel access sensitive military data.

6.2 Machine Learning Governance & Model Transparency

Provides complete transparency and real-time validation of the underlying AI microservice:

- **Model Architecture:** Random Forest Classifier / LightGBM Ensemble.
- **Genuine Evaluated Accuracy:** Verified at **81.5%** against test datasets.
- **Latency Benchmark:** Sub-50ms inference time per check-in assessment.
- **Inference Health:** Real-time ping test against the FastAPI ML microservice (`http://127.0.0.1:8000/health`).

6.3 System Health & Telemetry Dashboard

Monitors operational stability across all microservices:

- **Node.js API Server:** Uptime, active connections, memory usage, and route status.
- **FastAPI ML Microservice:** Microservice availability, endpoint response times.
- **Database Adapter:** Data persistence integrity, record count across collections.

6.4 Immutable Security Audit Trail Logs

Every sensitive action taken in the system is recorded in an immutable security audit log:

- **Fields Captured:** UTC Timestamp, Action Type (e.g., `USER_LOGIN`, `ALERT_REVIEWED`, `ROSTER_PACING_APPROVED`, `SUPPORT_RESOLVED`), User ID, Target Personnel ID, and IP Address.
- **Compliance & Accountability:** Ensures complete forensic traceability and prevents unauthorized tampering with medical or military duty records.

7. Core Algorithms & Mathematical Formulations

7.1 Perceived Stress Scale (PSS-10) Normalization

The PSS-10 assessment consists of 10 items scored on a 5-point Likert scale (0 = Never to 4 = Very Often). Items 4, 5, 7, and 8 are positively phrased and reverse-scored:

Reverse Score: $S_i = 4 - R_i$ (for items 4, 5, 7, 8) Raw PSS Score: $PSS_{raw} = \sum(S_i)$ from $i=1$ to 10 (Range: 0 to 40) Normalized PSS: $PSS_{norm} = (PSS_{raw} / 40) * 100$ (Range: 0 to 100%)

7.2 Composite Risk Score Formulation

The final welfare index combines psychometric scoring with physiological strain and operational load:

Composite Risk = $(0.45 * ML_Probability) + (0.25 * Normalized_PSS) + (0.20 * Sleep_Fatigue_Index) + (0.10 * Duty_Overload_Index)$

- **0.00 to 0.40 (0% - 40%):** **LOW / BALANCED BASELINE**
- **0.41 to 0.65 (41% - 65%):** **MODERATE STRAIN SIGNAL**
- **0.66 to 1.00 (66% - 100%):** **HIGH WELFARE CONCERN** (Triggers automated officer alert)

7.3 Trajectory Velocity Formula

Measures the rate of recovery or strain accumulation across consecutive check-ins:

Velocity (Delta) = $Current_Risk_Score - Previous_Risk_Score$ If Delta $\leq -6\%$: IMPROVING RECOVERY (Green) If Delta $\geq +6\%$: RISING STRAIN VELOCITY (Red) Otherwise: STABLE TRAJECTORY (Amber)

8. Judge Presentation & Evaluation Defense Checklist

When presenting this software to evaluation judges or defense leadership, use this structured walkthrough:

8.1 3-Minute Elevator Pitch

"Respected Judges, in high-stress defense and paramilitary environments, mental health strain remains invisible due to stigma and operational intensity until a tragic crisis occurs. WelfareAI transforms defense welfare from a reactive crisis response into an active, predictive operational radar. By combining clinically validated PSS-10 scales with physiological and duty telemetry, our machine learning ensemble detects compounding strain weeks early. Crucially, our system bridges clinical diagnosis with operational military command through the AI Shift Pacing Optimizer—allowing commanders to rebalance duty rosters, track closed-loop recovery velocities, and resolve confidential support requests with full audit integrity."

8.2 Live Demonstration Flow (Recommended Order)

- Soldier Check-in:** Log in as soldier (CRPF-9042). Open stress-checkin.html, complete an assessment showing poor sleep (4h) and high duty (56h).
- Instant ML Inference:** Show ai-analysis.html displaying the real-time gauge, factor breakdown, and explainability chart.
- Welfare Officer Command:** Log in as Officer. Show welfare-officer.html:
 - Show the new alert in **AI Triage & Alerts**.
 - Show the soldier mapped onto the **4-Quadrant Early-Warning Radar**.
 - Show the **Unit Personnel Roster** with live status badges.
 - Demonstrate the **AI Shift Pacing Optimizer**: approve the 72h daylight duty rotation and explain the predicted 34% relief.
 - Respond to a confidential counseling request in the **Support Requests Queue**.
- Soldier Recovery Verification:** Log back in as soldier. Show that the officer's resolution note appears in support.html and the notification bell rings.
- Admin Governance:** Log in as Admin. Show the 81.5% model accuracy transparency and the immutable security audit log capturing all previous actions.

8.3 Top 5 Expected Judge Questions & Best Answers

Question	Winning Defense Answer
1. "Will soldiers hesitate to use this due to stigma or career fear?"	"We designed WelfareAI with strict Non-Punitive Medical Safeguards. Individual psychometric scores are confidential. Welfare Officers see welfare signals, not personal diary entries. Furthermore, the system empowers soldiers to self-monitor and make micro-lifestyle adjustments before any command escalation is needed."

Question	Winning Defense Answer
2. "Is the AI a black box? How do you explain results?"	<i>"Not at all. We employ Explainable AI (XAI) feature attribution. Every prediction explicitly itemizes how much sleep, duty hours, and PSS questions contributed to the final score. The soldier and officer see the exact clinical rationale behind every recommendation."</i>
3. "How does this fit into real military duty rosters?"	<i>"Through our AI Shift Pacing Optimizer. Instead of simply saying a soldier is stressed, it generates formal Adjutant Protocols (e.g. PROP-CRPF-9042-801) proposing realistic duty reallocations (like swapping night watch for daylight logistics) to restore sleep homeostasis while maintaining unit readiness."</i>
4. "Does it work in remote field areas with low bandwidth?"	<i>"Yes. The frontend is built on pure lightweight Vanilla HTML5/CSS/JavaScript with zero heavy framework bloat. Pages load in under 200 milliseconds, and the architecture supports offline caching and asynchronous batch synchronization."</i>
5. "How do you ensure data security and prevent tampering?"	<i>"All data access is guarded by JWT authentication and strict Role-Based Access Control (RBAC). Every login, review, duty shift approval, and status change is immutably logged in our security audit trail with user identity, timestamp, and IP tracking."</i>